

The Same Layer in PyTorch, and on a GPU

PyTorch's arrays are called tensors, its notation is nearly NumPy's, and it can run the same line on a second processor.

From arrays to tensors

Name	Dimensions	Example in this meeting	Shape
scalar	0	the bias of one neuron	a single number
vector	1	one pixel; the weights of one neuron	(3,)
matrix	2	the points X; the weights W of a layer	(n, 3) and (K, 3)
volume	3	a colour image	(height, width, 3)
(no special name)	4	a batch of colour images	(30, 96, 96, 3)

- All of them: **tensors**. Here, a data structure: an array with some number of dimensions
- PyTorch's arrays are tensors: `torch.from_numpy`, `torch.tensor`

Three tensors

```
w_t = torch.tensor([0.299, 0.587, 0.114])    # the weights of the grey neuron, from a list
X_t = torch.from_numpy(X)                  # the points, from the NumPy array X
image_t = torch.from_numpy(image)          # the image, from the NumPy array image
```

```
w_t      dimensions: 1   shape: (3,)   type of its numbers: torch.float32
X_t      dimensions: 2   shape: (320640, 3)   type of its numbers: torch.float64
image_t  dimensions: 3   shape: (501, 640, 3)   type of its numbers: torch.float64
```

- `w_t`, made by PyTorch: `float32`, 4 bytes, about 7 digits
- `X_t`, from NumPy: kept `float64`, 8 bytes, about 16 digits
- `float32`: enough for networks, half the memory, fastest on a GPU

A first difference: the type of the numbers

```
try:
    X_t @ w_t
except RuntimeError as error:
    print("X_t @ w_t fails:", error)

X_t = X_t.float()                                # the same numbers, as float32
megabytes = X_t.element_size() * X_t.nelement() / 1e6
print("after X_t = X_t.float(), the type of its numbers is", X_t.dtype)
print(f"memory of the points: {X.nbytes / 1e6:.1f} MB as float64, {megabytes:.1f} MB as float32")

s_t = X_t @ w_t                                  # one neuron, many points: the line of Act 1
```

```
X_t @ w_t fails: addmv input tensors must have the same dtype, but got Float, Double, and Float
after X_t = X_t.float(), the type of its numbers is torch.float32
memory of the points: 7.7 MB as float64, 3.8 MB as float32
shape of X_t @ w_t: (320640,)
```

- Two types do not multiply: the error
- `.float()`: the same numbers as float32, half the memory

The layer of Act 1, in PyTorch

```
W_t = torch.from_numpy(W).float()      # the weights of the layer, as a float32 tensor
b_t = torch.from_numpy(b).float()      # the biases of the layer

S_t = X_t @ W_t.T + b_t                # the line of Act 1, unchanged
print("shape of S_t:", tuple(S_t.shape))
print(f"largest difference from the NumPy scores S: {np.abs(S_t.numpy() - S).max():.1e}")
```

```
shape of S_t: (320640, 5)
largest difference from the NumPy scores S: 1.8e-07
```

- `X @ W.T + b`: unchanged
- `@`, `.T`, broadcasting: as in NumPy
- Agreement with NumPy to about seven digits: float32

NumPy and PyTorch, side by side

	NumPy	PyTorch
make an array from a list	<code>np.array([1.0, 2.0])</code>	<code>torch.tensor([1.0, 2.0])</code>
the layer	<code>X @ W.T + b</code>	<code>X @ W.T + b</code>
a function of every entry	<code>np.exp(S)</code>	<code>torch.exp(S)</code>
along the rows	<code>axis=1, keepdims=True</code>	<code>dim=1, keepdim=True</code> ; NumPy's names are accepted too
the largest entry of each row	<code>S.max(axis=1)</code> is an array	<code>S.max(dim=1)</code> holds <code>.values</code> and <code>.indices</code>
decimal numbers	<code>float64</code>	<code>float32</code>
where the numbers are	in the memory of the CPU	on a device : the CPU or a GPU

The softmax, translated

```
def softmax_rows(S):  
    shifted = S - S.max(axis=1, keepdims=True)  
    E = np.exp(shifted)  
    return E / E.sum(axis=1, keepdims=True)
```

```
def softmax_rows_torch(S):  
    shifted = S - S.max(dim=1, keepdim=True).values  
    E = torch.exp(shifted)  
    return E / E.sum(dim=1, keepdim=True)
```

- Translated by an AI assistant to stated requirements: a tensor in, a tensor out, no loop, no NumPy
- Three differences: `torch.exp`; `dim` and `keepdim`; `.values`
- No NumPy inside: NumPy cannot compute with a tensor on a GPU

PyTorch's own layer

```
linear = torch.nn.Linear(3, K)           # a layer of K neurons for points with 3 coordinates
print("shape of linear.weight:", tuple(linear.weight.shape))
print("shape of linear.bias: ", tuple(linear.bias.shape))

with torch.no_grad():                    # this cell needs no gradients
    linear.weight.copy_(W_t)              # the weights of Act 1 go into the layer
    linear.bias.copy_(b_t)                # and so do the biases
    S_linear = linear(X_t)                # the layer, applied to all the points
```

```
shape of linear(X_t): (320640, 5)
```

```
largest difference between linear(X_t) and X_t @ W_t.T + b_t: 0.0e+00
```

- `torch.nn.Linear(3, K)`: K neurons for points with 3 coordinates
- Weights of shape (K, 3): **one neuron in each row**. $y = xA^T + b$
- `torch.no_grad()`: this cell needs no gradients

A second processor: the GPU

- **CPU**, the central processing unit: a few **cores**, each fast, any program
- **GPU**, the graphics processing unit: thousands of small cores, the same instruction on many numbers
- Its own memory. The T4 that Colab lends: 2,560 cores, 16 GB, 8.1 trillion operations a second at peak
- Built for the pixels of a screen: one short computation, millions of times
- A product of matrices: work of the same kind

The page **CPU and GPU**, parts 1 and 2: ikoutis.github.io/cs301-labs/unit-2/pages/cpu-and-gpu.html

A tensor lives on a device

```
if GPU:
    X_g = X_t.to("cuda")           # copies of the three tensors, in the memory of the GPU
    W_g = W_t.to("cuda")
    b_g = b_t.to("cuda")
    print("X_g is on the device:", X_g.device)

    try:
        X_g @ W_t.T                 # X_g is on the GPU and W_t is on the CPU
    except RuntimeError as error:
        print("X_g @ W_t.T fails:", error)

    S_g = X_g @ W_g.T + b_g         # the line of Act 1, carried out by the GPU
    print("S_g is on the device:", S_g.device)
```

- `X_t.to("cuda")`: a **copy** in the memory of the GPU
- One operation, one device: `X_g @ W_t.T` fails. `.cpu()` copies back
- The line of the scores: unchanged

Measure the product on both processors

```
def time_product(size, device):  
    """The average time, in seconds, of X @ W.T + b on the device, with n = m = K = size."""  
    X = torch.rand(size, size, device=device) # random float32 numbers, made on the device  
    W = torch.rand(size, size, device=device)  
    b = torch.rand(size, device=device)  
    repeats = max(1, min(200, 8_000_000_000 // size ** 3)) # many repeats when the product is small  
    finish(device)  
    start = time.perf_counter()  
    for _ in range(repeats):  
        S = X @ W.T + b  
    finish(device)  
    return (time.perf_counter() - start) / repeats
```

- `n = m = K = size`; random numbers made on the device; many repeats when small
- `torch.cuda.synchronize()`: wait for the GPU before reading the clock
- One small product first, not measured: the first use of a device is slow

The time of the product on a CPU

`X @ W.T + b` with $n = m = K = \text{size}$, in seconds. CPU: 32 logical cores. GPU: none.

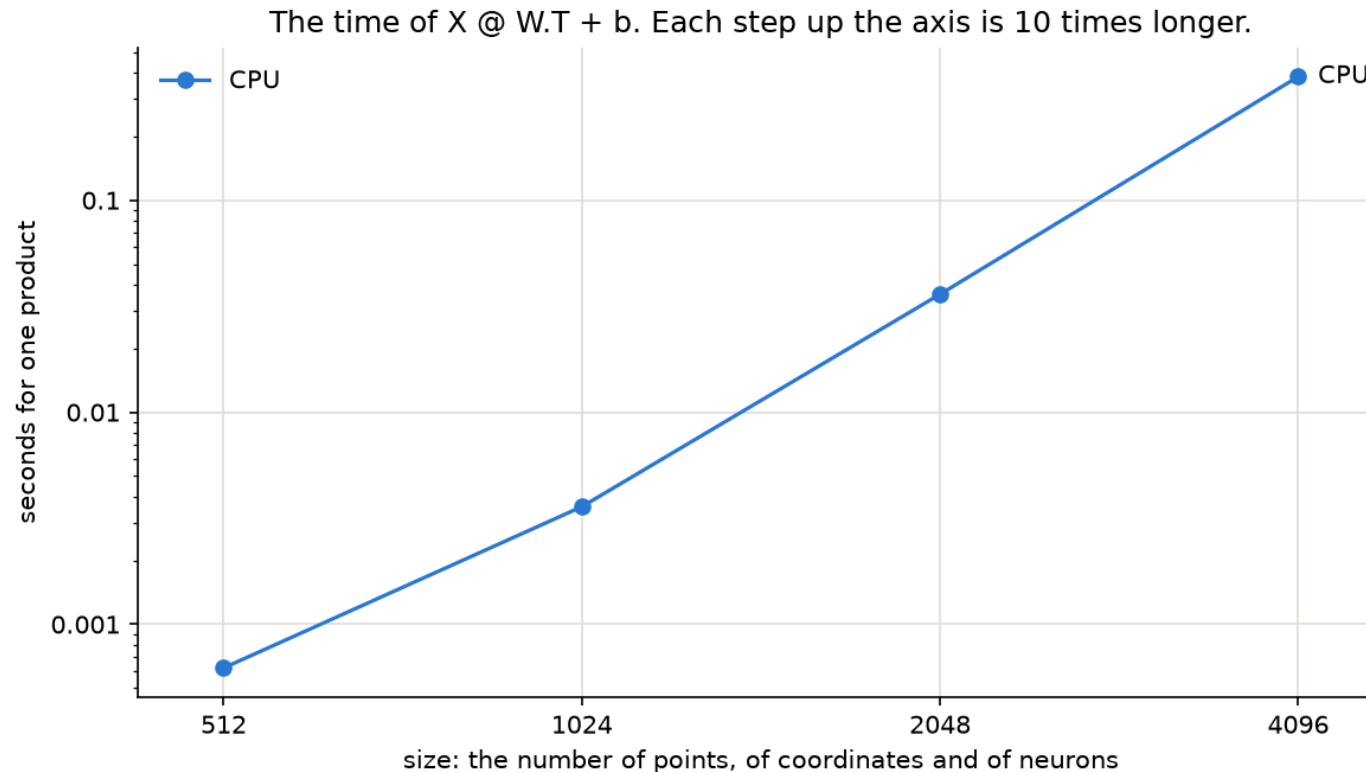
size	multiplications	CPU
512	134,217,728	0.000623
1024	1,073,741,824	0.003585
2048	8,589,934,592	0.036016
4096	68,719,476,736	0.382494

- Measured here, on a CPU with 32 cores. 2,048 $\rightarrow$ 4,096: the time grows by a factor near 8
- What to expect from Colab's T4 at size 4,096, from published measurements:

	operations a second	the product at size 4,096
Colab T4, cuBLAS, measured by others	4.6 trillion	about 0.030 s
T4 at its datasheet peak	8.1 trillion	0.017 s

- **Your GPU column is your own measurement.** Expect the GPU to be faster at every size here, tens of times at 4,096

The same numbers as a chart



- Both axes stretched: each size $\times 2$, each mark on the vertical axis $\times 10$
- A straight line: the same factor for each doubling. Yours will have a second line

Now in Colab: Act 2

- **Act 2** of the class notebook: the layer in PyTorch, and on the GPU if your session has one
- Run the measurement: four sizes, 512 to 4,096, both devices, up to half a minute
- From the table: by what factor is the GPU faster at 4,096? By what factor does the CPU's time grow from 2,048 to 4,096?
- **Post**, as a reply to your first post: the line, the chart with one sentence, your two answers, your GPU's name
- One member posts; the names of all the members first